

**Institute of Mathematical Sciences
University of Malaya**

**SIM1001 Basic Mathematics
SIX1013 Fundamentals of Advanced Mathematics**

TUTORIAL 5 (PART 1)

1. Find the remainder when $x^3 + 3x^2 - x + 2$ is divided by
 - (a) $x + 3$
 - (b) $3x - 1$
 - (c) $x - 2 - i$
 - (d) $x^2 - 2x - 3$
 - (e) $x^2 - 4x + 4$
2. By using the long division, find the quotient and the remainder when the polynomial $P(x)$ is divided by $D(x)$ in each of the following cases:
 - (a) $P(x) = 4x^4 - 6x^2 + 8x + 1$, $D(x) = 2x^2 - 5$
 - (b) $P(x) = x^7 + 5x^6 - 4x^3 + 2x^2 + 1$, $D(x) = x^4 - 1$
3. By using the synthetic division, find the quotient and the remainder when the polynomial $2x^4 + 3x^3 + 2x^2 + x - 1$ is divided by
 - (a) $x - 3$;
 - (b) $2x - 1$;
 - (c) $2x^2 + 5x - 3$.
4.
 - (a) Verify that 2 is a zero of $P(x) = x^3 - 7x^2 + 12x - 4$. Hence find all zeros of $P(x)$.
 - (b) Solve the equation $2x^4 - 9x^3 + 6x^2 + 20x - 24 = 0$ given that it has a root of multiplicity 3.
5.
 - (a) Show that $x - 1$ is a factor of the polynomial $2x^3 - 9x^2 + 3x + 4$ and factorize this polynomial completely.
 - (b) Use the result in part (a) to obtain all linear factors of the polynomial $2x^6 - 9x^4 + 3x^2 + 4$.
6.
 - (a) Verify that $5 + i$ is a zero (root) of the polynomial
$$P(x) = x^4 - 10x^3 + 24x^2 + 20x - 52$$
and factorize $P(x)$ completely.
 - (b) Show that $1 + i$ is a zero (root) of the polynomial $P(x) = x^2 + (1 - i)x - 2(1 + i)$. Does its conjugate also a zero of $P(x)$?

7. Find the real polynomial $P(x)$ of degree 7 with the following properties:
- (i) The leading coefficient of $P(x)$ is 1,
 - (ii) i is a multiple root of $P(x)$,
 - (iii) $2i$ is a root of $P(x)$, and
 - (iv) the remainder in the division of $P(x)$ by $x - 1$ is 40.
8. (a) List all the possible rational roots of the equation $3x^3 - x^2 + 2x + 20 = 0$.
Is $-\frac{5}{3}$ one of its roots?
- (b) Find all the linear factors of the polynomial $2x^4 - 5x^3 + 6x^2 - 10x + 4$.
9. When m is a positive integer ≥ 2 , show that $x^m + 2x^{m-1} - 2$ has no rational root.
10. Let α_1, α_2 and α_3 be all the roots of the equation $3x^3 + 4x^2 + 5x + 6 = 0$.
- (a) Show that $\alpha_1 + \alpha_2 + \alpha_3 = -\frac{4}{3}$ and $\alpha_1\alpha_2\alpha_3 = -2$.
 - (b) Show that the equation has a real root.